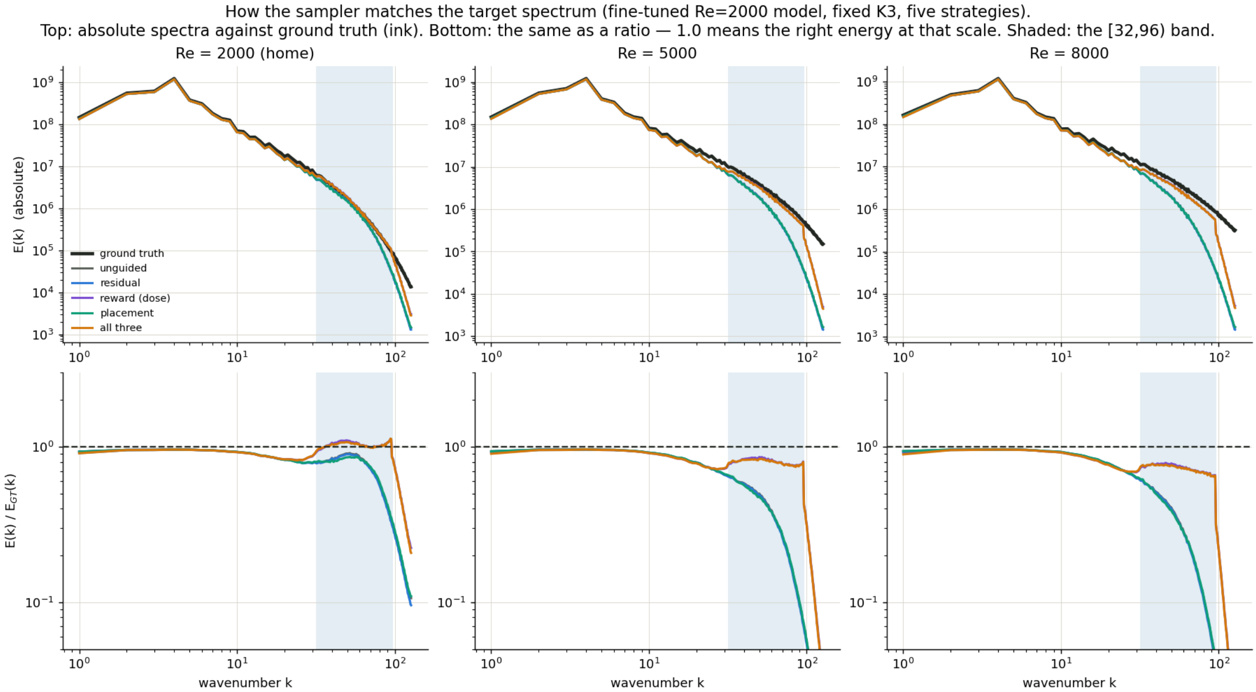
The Steering Dial: reward-directed Langevin sampling for cross-regime super-resolution

Differentiable statistics gradients composed into the diffusion sampler — the complete record: two potentials, measured targets, the full strategy grid, blind auto-calibration, trajectories. All Track-A (measured statistics, no paired fine samples). 2026-08-23.

Verdicts

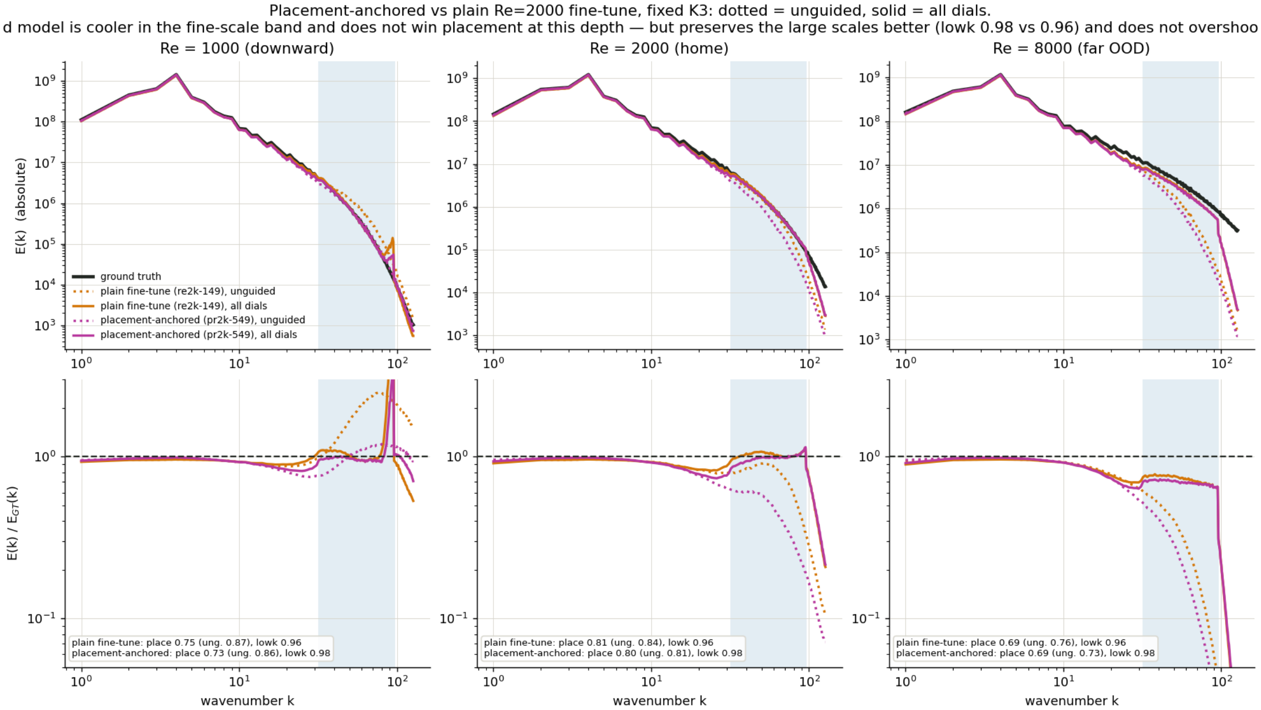
(1) The spectral (reward) gradient is a true bidirectional dose dial: monotone, self-limiting at measured targets, heats cold models OOD and cools hot ones downward, and is the only sampling strategy that carries fine-scale energy across regimes. **(2)** The physics (residual) gradient is a polisher: flat on retention and placement everywhere, lowest residual — the historical λ=3 guidance was a near no-op. **(3)** The placement gradient buys +0.02–0.04 placement essentially free and refunds ~40% of the dose dial's placement cost at fixed dose (saturating — the ceiling is the observation's information content). **(4)** The dials are additive: three independent one-dimensional tunings. **(5)** The dial tunes itself blind: stat-match auto-calibration holds retention within ±11% of truth at all nine regimes. **(6)** Steered-K3 beats plain K4 at Re=8000 (0.846 vs 0.750 at 86 vs 110 network passes): the dial replaces cascade rungs.

0c · The object of the exercise: matching the energy spectrum



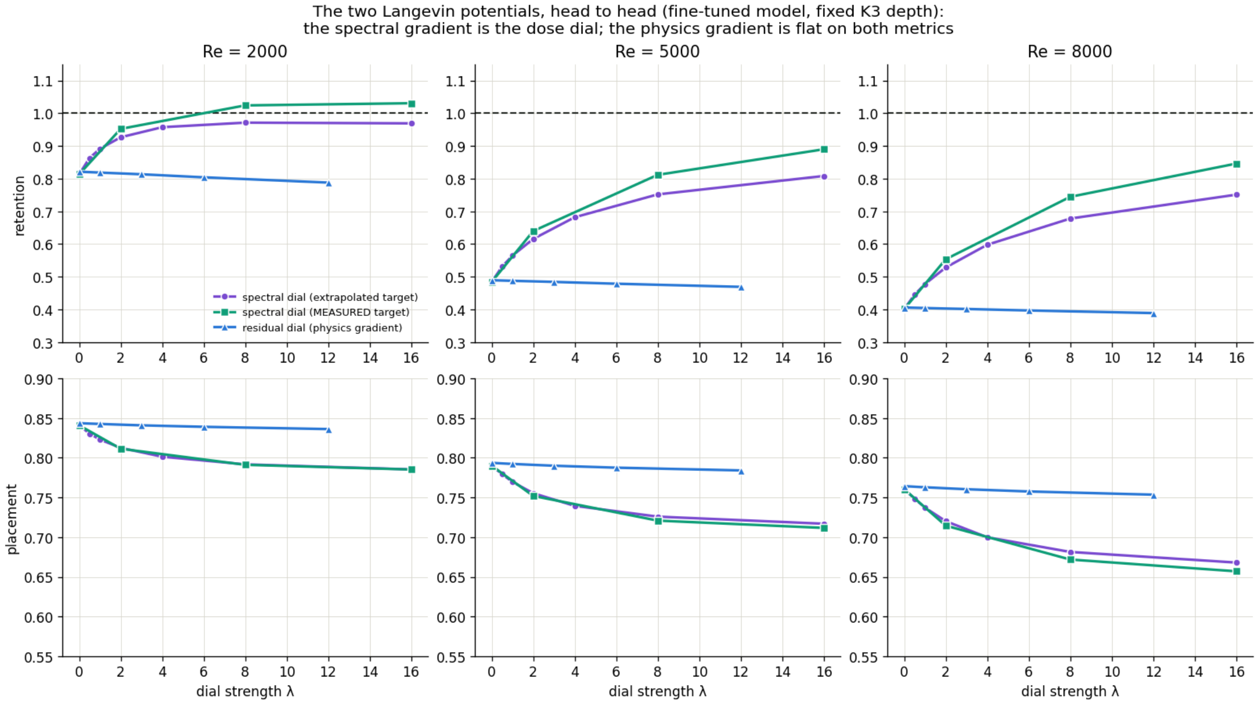
The same data two ways (fine-tuned Re=2000 model, fixed K3, five strategies). Top: absolute spectra — every strategy hugs the ground-truth line (ink) until the shaded [32,96) band, where the unguided/residual/placement bundle peels below it while the dose-guided curves stay pinned to the band edge. Bottom: the ratio view the rest of this report uses — that barely-visible peel-off is a factor-of-10 deficit at the deepest scales. Matching the spectrum is easy; matching the last two octaves is the entire game.

0d · Placement-anchored vs plain fine-tune, spectrally

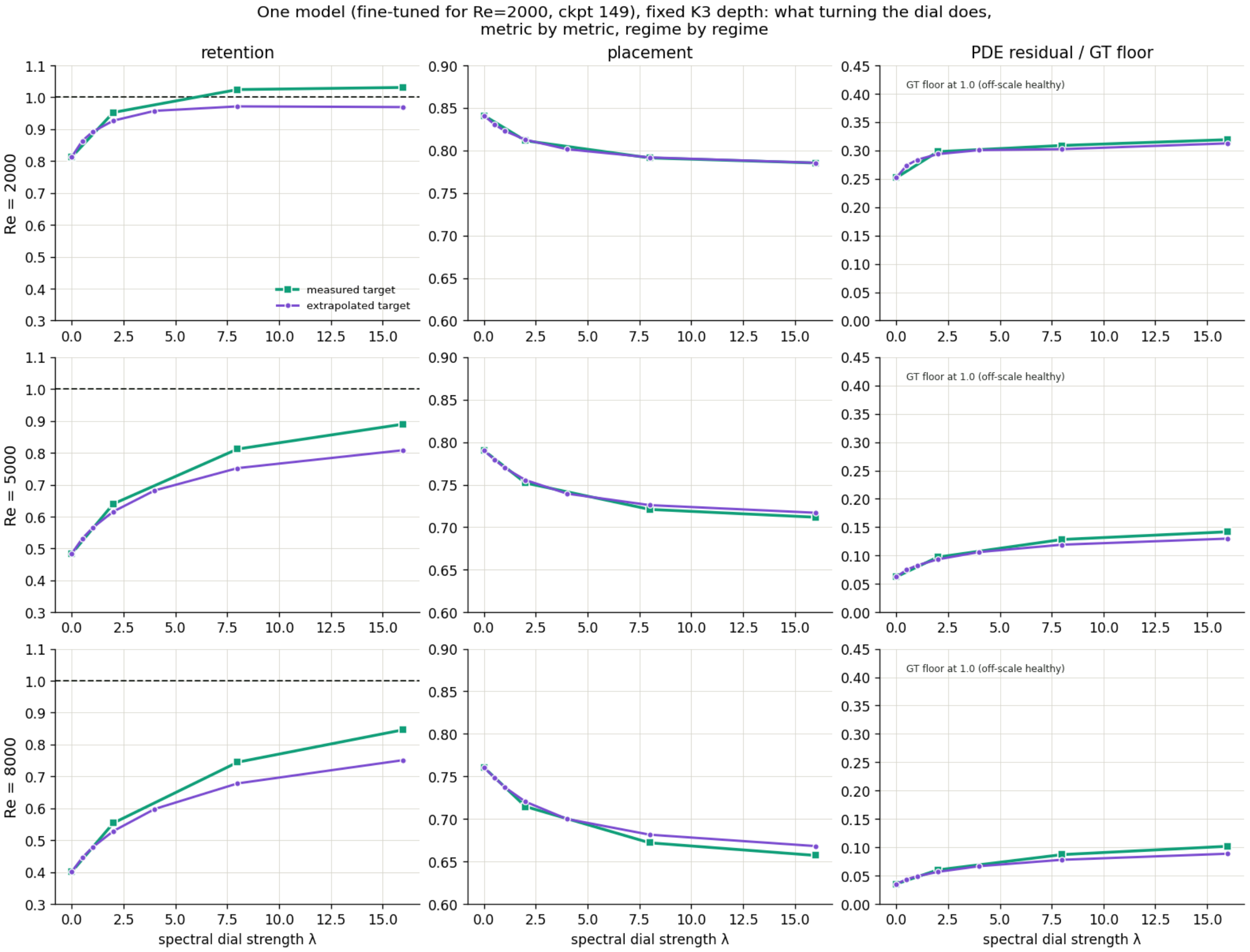


Fixed K3 head-to-head: the placement-anchored model runs cooler in the fine-scale band (unguided 0.57 vs 0.82 at home) and its placement advantage (a matched-deep-depth property from the cards) does not survive at K3 (0.81 vs 0.84). It keeps the healthiest large scales in the family (lowk 0.98 vs 0.96) and never overshoots downward (0.88 vs 1.33 at Re=1000). A regularized, conservative variant: for a fixed shallow config, plain fine-tune + dials dominates.

1 / 1b · Two potentials, metric by metric

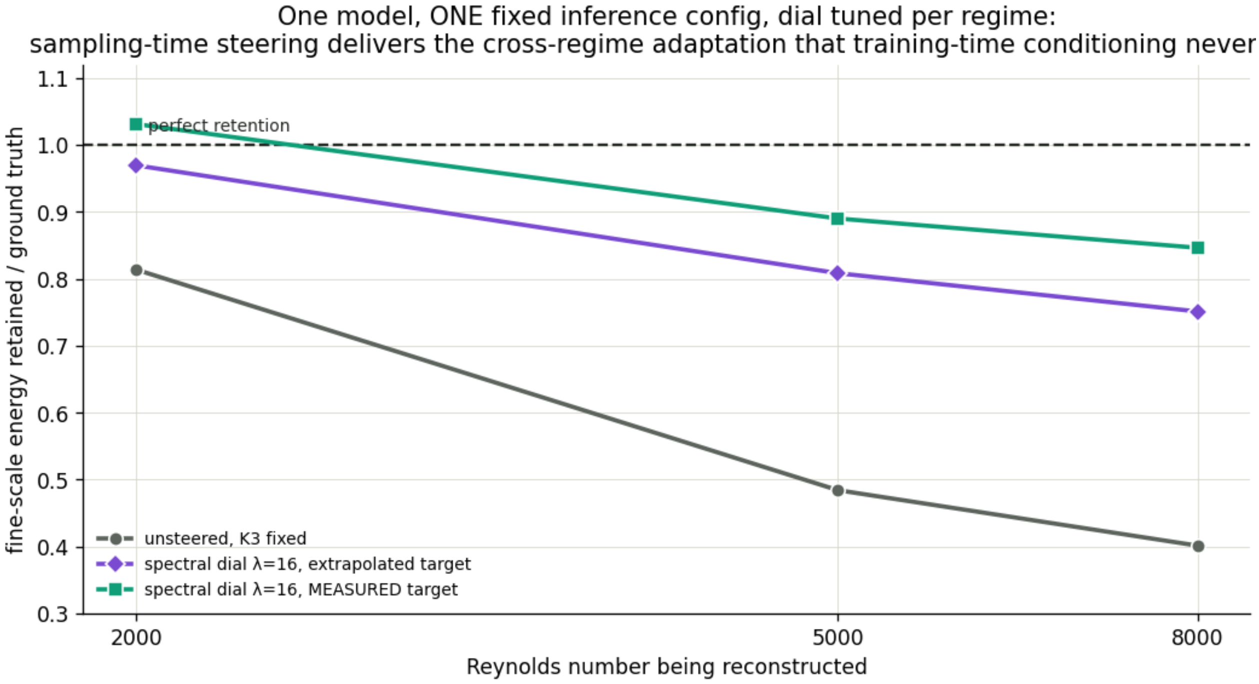


Retention and placement vs dial strength for the spectral dial (extrapolated and measured targets) and the residual dial, re2k-149 @ K3, Re 2000/5000/8000.



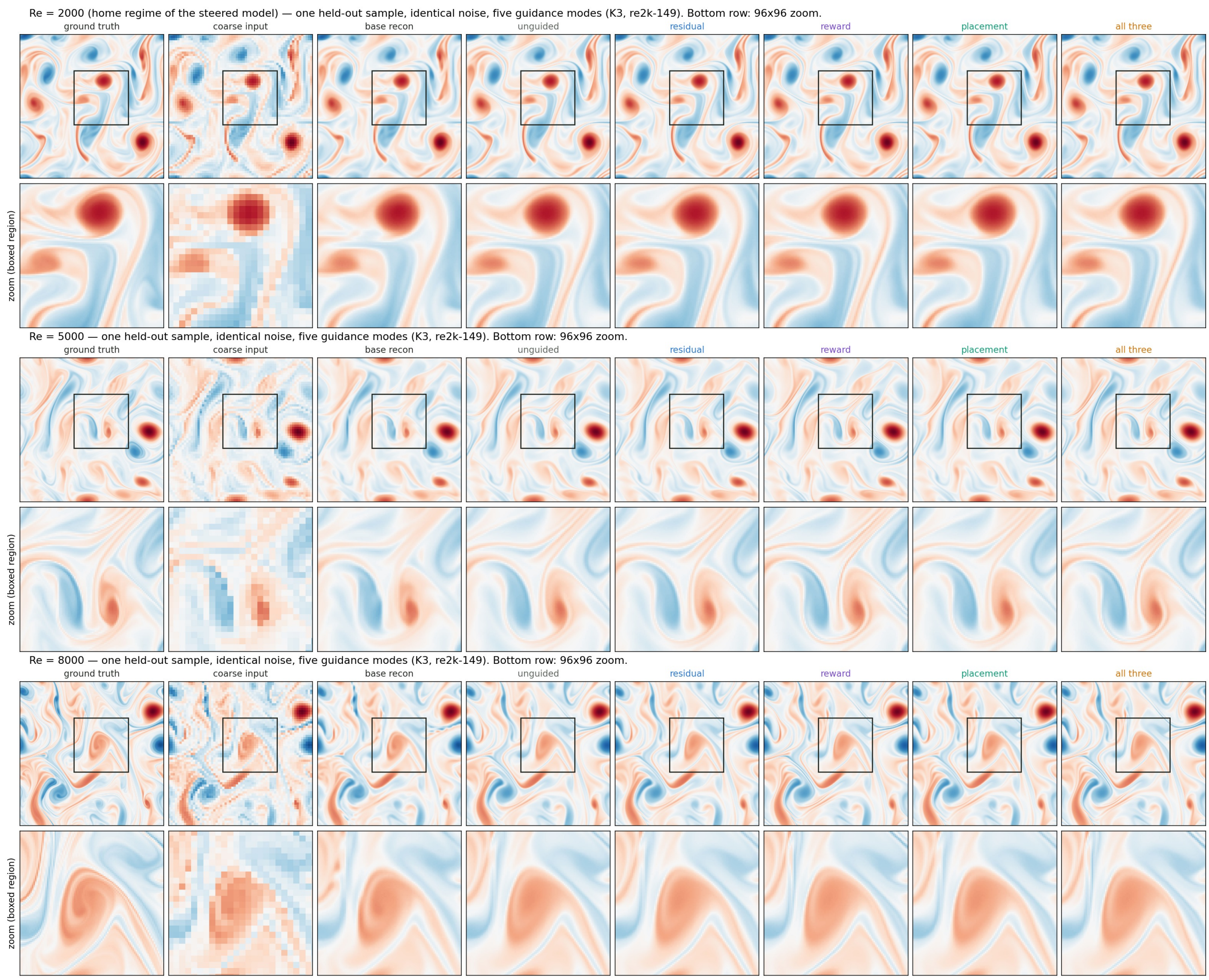
All three metrics per regime: retention climbs, placement pays 0.05–0.10, residual rises with injected energy yet stays 3–10x below the GT floor.

2 · One fixed configuration across regimes



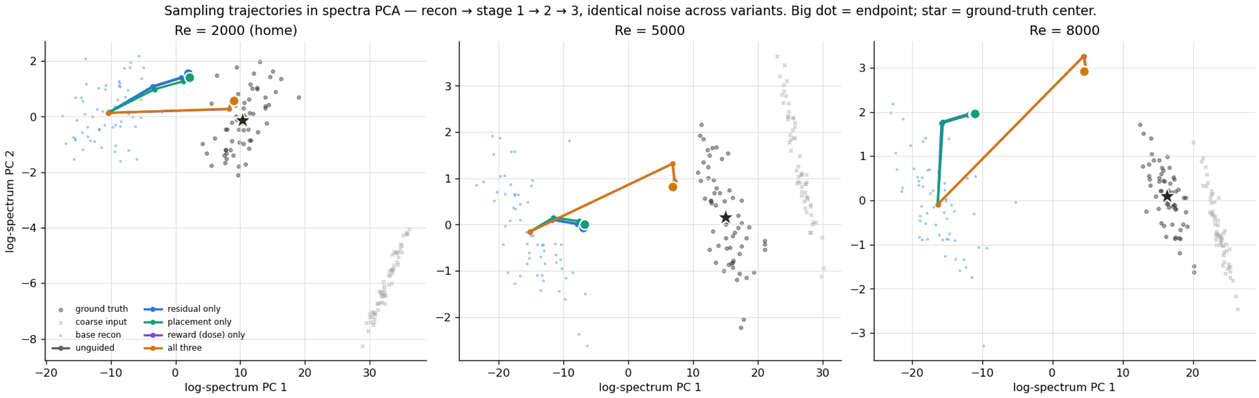
The cross-regime adaptation training-time conditioning never delivered: unsteered 0.81/0.48/0.40 becomes 1.03/0.89/0.85 with measured targets, one config.

2b · Renders: what the dials look like

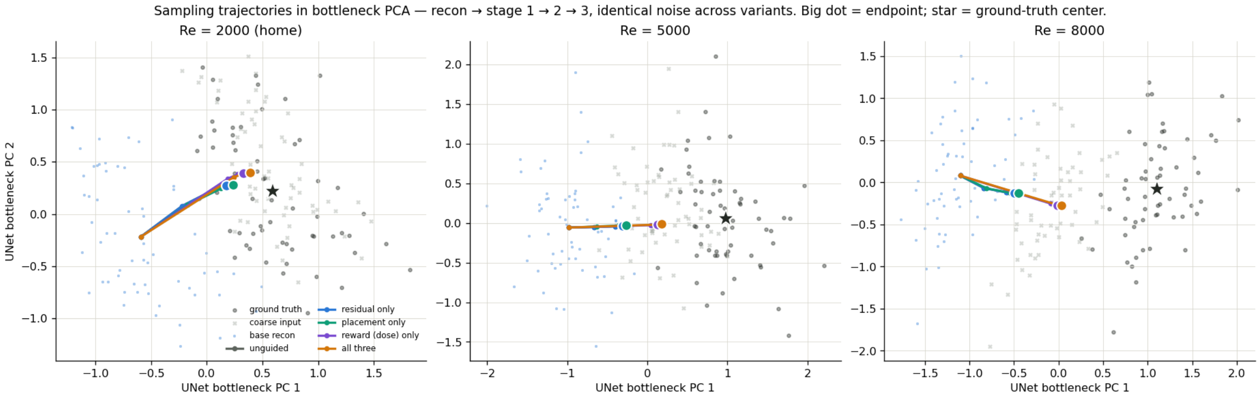


One held-out sample per regime, identical noise in every panel; bottom rows are 96x96 zooms. Residual is indistinguishable from unguided; reward and all-three restore the filamentary banding the recon washes out.

2b (cont.) · Sampling trajectories in embedding space



Log-spectrum PCA: unguided/residual/placement bundle and stall; the dose path crosses the gap, landing ON the GT cloud at home — endpoint separation grows with regime distance. The violet reward curve hides under orange all-three: additivity, drawn.



UNet bottleneck PCA: the regime clouds barely separate — the network's internal representation scarcely encodes regime, which is why training-time conditioning had no pathway to recruit and the correction must be imposed spectrally at sampling time.

2b2–2b4 · What fine-tuning buys that steering cannot; the band edge, closed; the seed floor

Re=1000, fixed K3	retention	placement	MSE
base model, unguided	0.48	0.92	0.39
base model + dose dial	0.93	0.76	0.42
fine-tuned r1k-449, unguided	1.02	0.87	0.41

In-regime, the dial lifts the untrained base from 0.48 to 0.93 retention with no training at all — but placement crashes 0.92 to 0.76, while the fine-tune delivers 1.02 AT 0.87: steering injects energy; only training placed it. Separately, the band-edge pilot closed the k=96 question: an inside taper (never consulting the reference past 96) left in-band metrics identical but LOWERED the free spillover into [96,120) and k — the hard edge stays, now by measurement. And thirty headline cells re-run under two extra seeds give max retention spread 0.007 (median 0.001): every claimed effect is 10-100x the noise floor.*

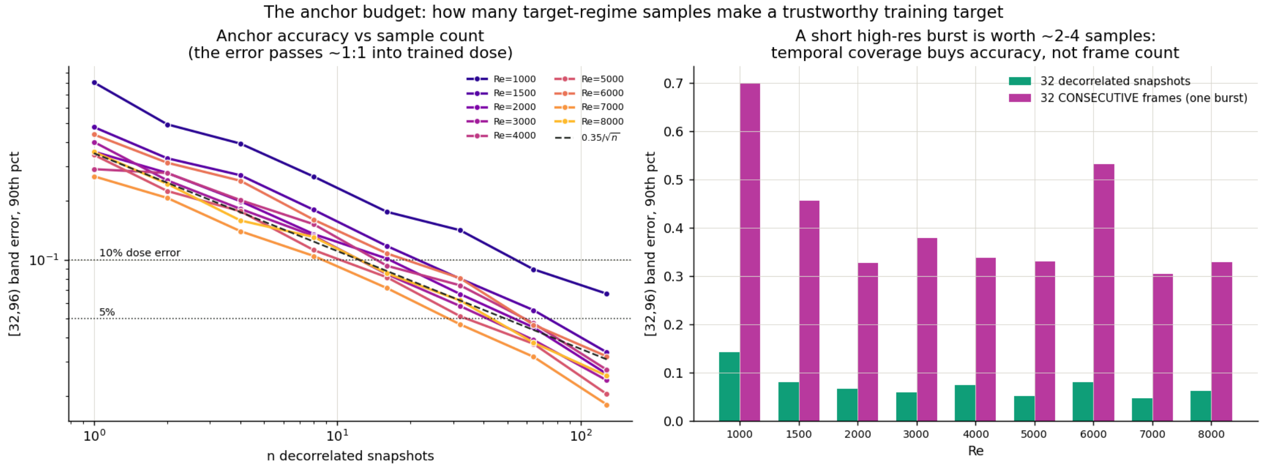
2b4b · The recon step across regimes

Re	recon ret [32,96)	dPlace [32,64)	dPlace [64,96)
1000	0.410	+0.050	-0.096
2000	0.308	+0.023	-0.088
4000	0.248	+0.008	-0.119
5000	0.232	-0.005	-0.175
8000	0.211	-0.007	-0.225

band (GT share @8000)	E_LR/E_GT @8000	E_recon/E_GT @8000	place LR->recon
[1,5) (53%)	0.99	0.99	preserved
[5,16) (34%)	0.96	0.95	preserved
[16,32) (7.9%)	0.90	0.71	preserved (0.95+)
[32,64) (4.2%)	1.19 (aliased)	0.26	tie
[64,96) (1.1%)	1.52 (aliased)	0.04	-0.23

Full accounting (Re=8000 shown; 2000/5000 in the web version): the recon's only positional cost is the [64,96) band (0.2-1.1% of energy, loss growing with Re), largely recovered already by the production 2-band dial (deep-band 0.52/0.44 at 5000/8000 via scale coupling), the 3-band dial adding +0.03/+0.08 on top; it also under-fills the real sub-Nyquist [16,32) band by 10-20%, which the cascade/dose dial refills. CONCLUSION: keeping the DDIM recon as the fine-tuning contract is well-supported - on-manifold initialization at an information price that is energetically negligible, positionally confined, and reclaimable. Raw-LR-init fine-tuning is unmotivated on this accounting.

2b5 · The anchor budget: how many samples make a trustworthy target



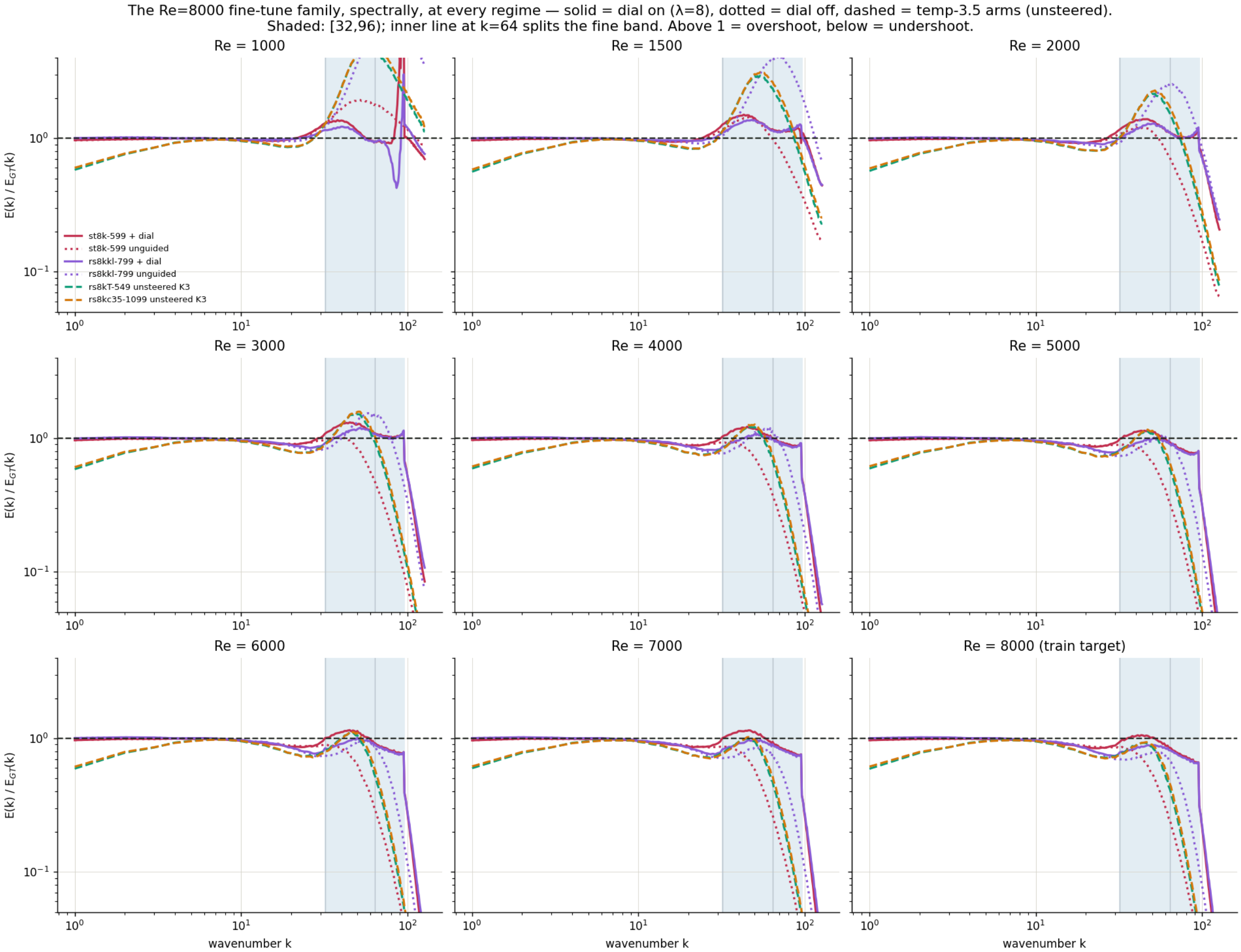
Bootstrap over the GT pools: a single snapshot mis-estimates the fine-scale dose by 27-81% (p90) — one triplet is NOT enough to fine-tune on. 16-32 decorrelated snapshots buy 10%-grade anchors (64 at Re=1000), ~64 buy 5%, converging as 1/sqrt(n). A consecutive burst is worth ~2-4 independent samples regardless of length; the vorticity distribution converges far faster than the spectrum. The sequence is the real unit (one DNS run): its slow envelope correlates its frames, so 1 sequence still carries 16-40% dose error, 2 give 10-21%, 4 give 6-13% - FOUR independent sequences buy a 10%-grade anchor. Economics: supervised retraining used ~36 paired sequences; the statistics route needs ~4 plus a 3.5 TPU-h PPO fine-tune - a ~9x saving in the dominant (DNS) cost, and below ~10 sequences the only viable route at all.

2b6 · Steered training: baking the dial into the weights

Re (dial on, lam=8, K3)	1000	2000	3000	4000	5000	6000	7000	8000
steered-trained st8k-599	1.27	1.26	1.19	1.10	1.03	1.03	1.03	0.95
conventional rs8kkl-799	1.14	1.14	1.06	0.96	0.90	0.90	0.89	0.82

The Re=8000 recipe retrained with the dose dial inside its rollouts (guidance in rollout AND loss, ratios exact; KL 0.003). It works: the flattest single-config cross-regime retention row in the program (1.03+-0.08 from Re=3000-8000), lowk 0.99 everywhere, placement matching the twin. Costs as predicted: dial-dependence (unguided 0.56 at home, best unguided placement 0.75) and low-Re overshoot at fixed lambda, trimmed by auto-lambda. If the dial is always on, this plus auto-lambda* is the strongest single-model single-config system measured here.*

2b7 · The steered family metric by metric, and retention over ALL bands



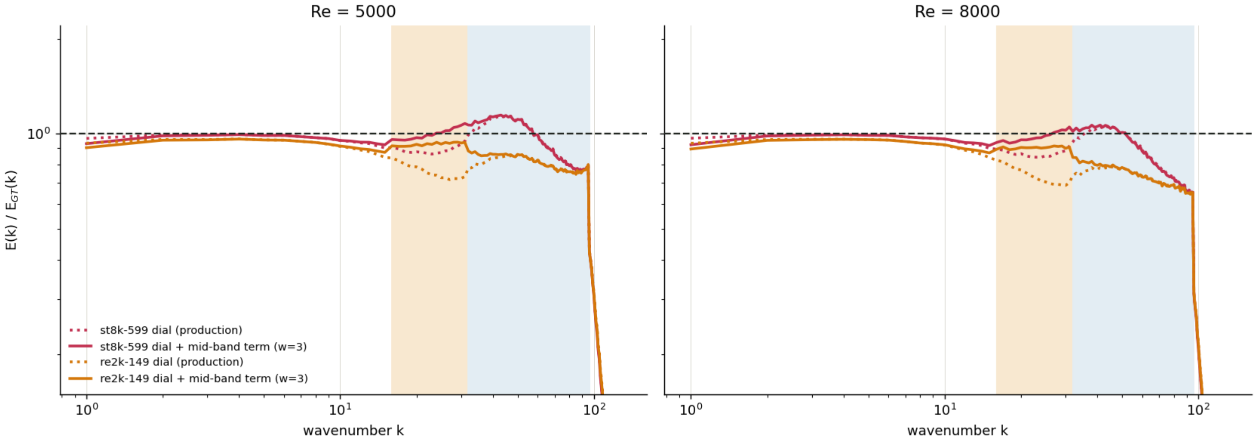
The Re=8000 family at every regime: the temp-3.5 arms start at 0.6 at k=1 everywhere (large-scale damage visible, disqualifying); st8k+dial rides the line through [32,64) at home and overshoots as a broad ~1.3 bump at Re<=2000; the mid-band sag is the shallow trough at k~20-30 in every OOD panel; and the far-downward corner shows the two dial artifacts (rs8kkl notch at k~90, st8k beyond-band spillover).

deployment mode (lam=8, K3)	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)
st8k-599 @ 1000	0.99	0.98	1.02	1.27	1.23
st8k-599 @ 8000	0.99	0.97	0.87	1.00	0.75
rs8kkl-799 @ 8000	1.01	0.99	0.82	0.84	0.73
re2k-149 @ 8000	0.96	0.94	0.75	0.76	0.69
base K3 @ 8000 (ref)	-	-	0.62	-	-

Per-band retention from every cell's stored spectrum. What aggregates hid: st8k's 0.95 at Re=8000 is a PERFECT 1.00 in [32,64) with the shortfall entirely in the deep tail; the [16,32) mid-band sag OOD (0.87/0.82/0.75, vs base 0.62 - already lifted +0.25 by fine-tune+dial) is the dose loss's 7x under-weighting of that band, which is fully observed (below input Nyquist) and near-perfectly placed - the easiest remaining win; the mid-band dial term test is queued. Aggressive-dial stability: raising lam *walks hot cells toward 1 at Re>=2000 (1.26 to 1.09 at lam=32) but the steered-TRAINED model has a stability ceiling at the far-downward extreme (runaway at Re=1000, lam>=16: ret 6-12, placement destroyed) - the auto-lambda* rule inherently rejects those settings (its meter reads the blowup).*

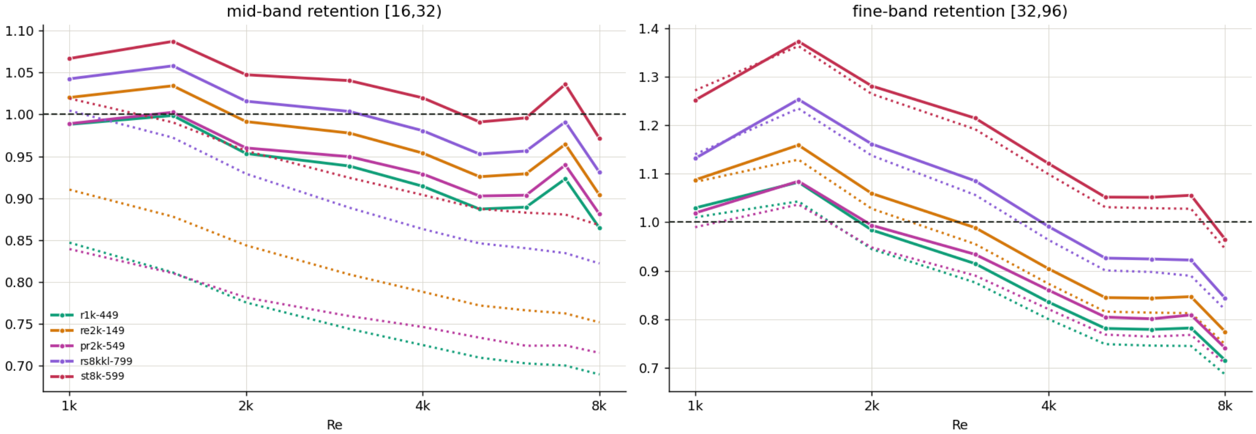
2b8 · The mid-band dial term: the sag closed at sampling time

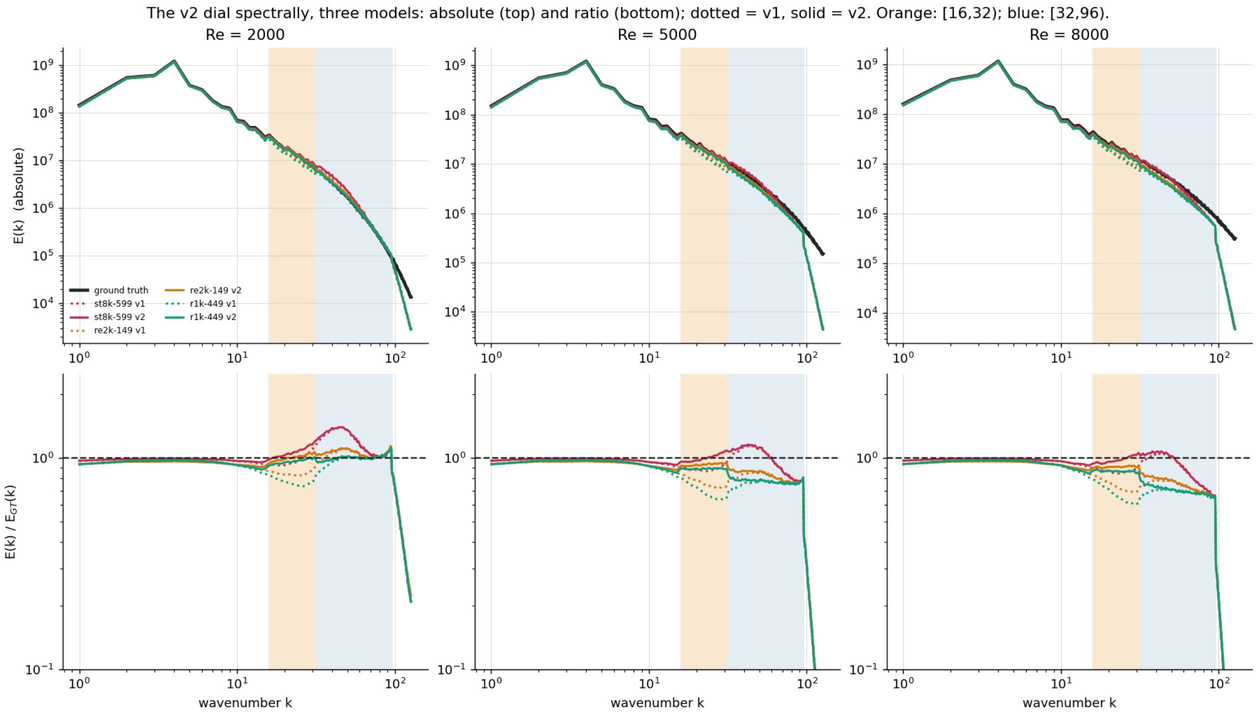
What the mid-band dial term does: dotted = production dose dial (λ=8), solid = with the added [16,32) term (w=3).
Orange shading: the [16,32) band it targets; blue: [32,96).



[16,32) retention	production dial	+ mid w=1.5	+ mid w=3
st8k-599 @ 5000	0.89	0.96	0.99
st8k-599 @ 8000	0.87	0.94	0.97
re2k-149 @ 5000	0.77	0.88	0.92
re2k-149 @ 8000	0.75	0.85	0.90

The v2 dose dial (added mid-band term, w=3) across the whole family and all regimes: dotted = production v1, solid = v2.





Re	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place
1000	0.99	0.98	1.07 (+0.05)	1.26 (-0.02)	1.19 (-0.05)	0.71
2000	0.99	0.98	1.05 (+0.09)	1.31 (+0.02)	1.05 (0.00)	0.78
4000	0.99	0.98	1.02 (+0.12)	1.16 (+0.03)	0.93 (0.00)	0.73
6000	0.99	0.98	1.00 (+0.11)	1.10 (+0.03)	0.84 (0.00)	0.72
8000	0.99	0.97	0.97 (+0.10)	1.02 (+0.02)	0.75 (0.00)	0.67

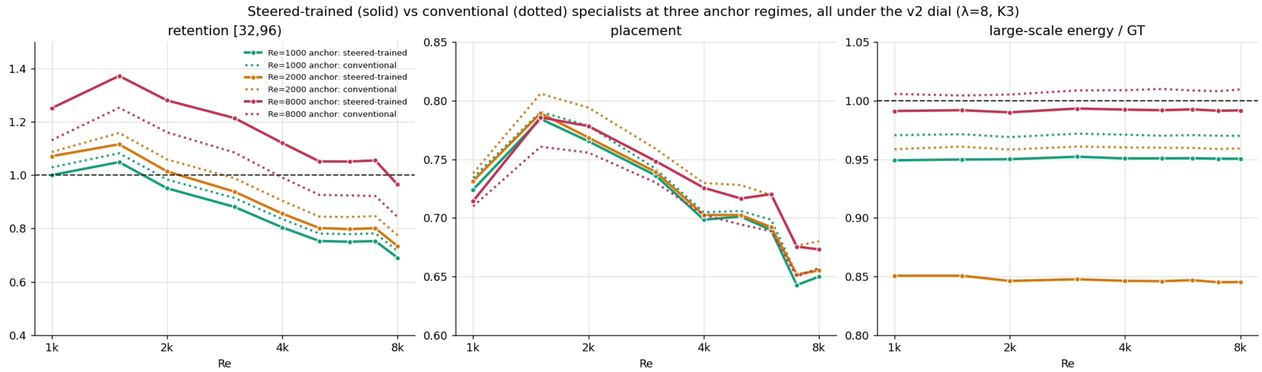
The deployment flagship (st8k-599 + v2 dial), per band per regime, deltas vs v1 in parentheses. Resolved scales pinned 0.97-0.99; mid band 0.97-1.09 everywhere; [32,64) mildly hot mid-transfer (per-regime calibration target); the deep tail is the one remaining OOD deficit (0.75-0.84, untouched by v2 as designed); placement unchanged. Spectra above: absolute (top) and ratio (bottom), v1 dotted vs v2 solid.

*The under-weighted [16,32) band closed by a third dose-loss term at sampling time - no fine-tuning. The full-family battery (45 cells: five models x nine regimes, same seeds) confirms: the mid band closes for every model everywhere (0.86-1.09, coldest models +0.15-0.22), collateral is a +0.03 average BONUS in [32,64) with placement unchanged, and at Re=8000 headline retention rises +0.02-0.03 (st8k reaches 0.97 at home). Production dial: 0.5*d[1,96) + 3*d_mid[16,32) + 3*d_hi[32,96). Mild new mid-band overshoot for hot models at Re<=2000 (1.02-1.09, fixed w=3) - per-regime weight selection would trim.*

championship @8000 (v2)	**ret**	**place**	**mid**	**tail**
st8k-599 (steered-trained)	0.965	0.673	0.97	0.75
rs8kkl-799 (conventional)	0.843	0.657	0.93	0.73
re2k-149 (transported)	0.775	0.680	0.90	0.69
championship @2000 (v2)	**ret**	**place**	**mid**	**tail**
pr2k-549	0.994	0.762	0.96	1.01
r1k-449	0.984	0.779	0.95	1.01
re2k-149	1.059	0.794	0.99	1.02

The championship: at Re=2000 the v2 dial equalizes the family (three models within +-6% of perfect; pr2k closest, re2k best-placed) - model choice barely matters near home. At Re=8000 the steered-trained specialist keeps +0.12 over its conventional twin and +0.19 over the transported model. Full per-model per-band tables (incl r1k-449) in the web version; the spectra figure above now carries all three flagship models.

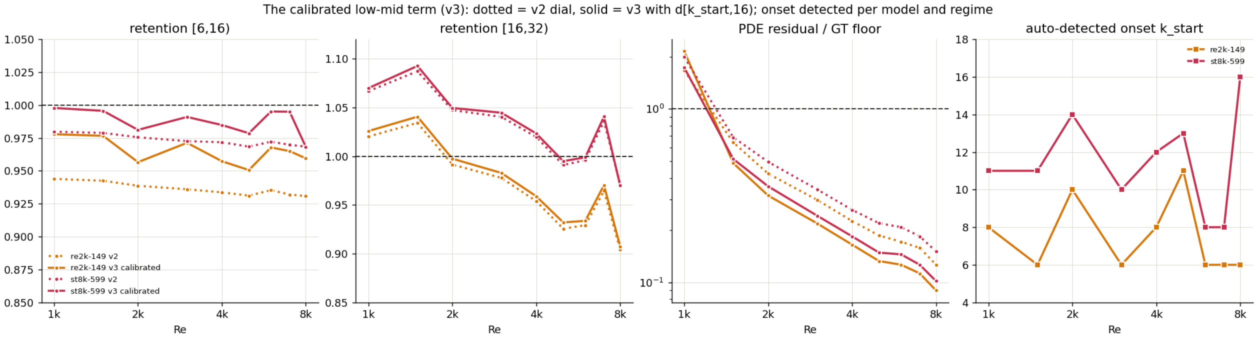
2b9 · Steered training at three anchors: where it pays, where it damages



home cell, v2 dial	ret	place	low-k	dial OFF ret
Re=1000 steered st1k / conv r1k	1.00 / 1.03	0.72 / 0.73	0.95 / 0.97	0.53 / 1.02
Re=2000 steered st2k / conv re2k	1.02 / 1.06	0.77 / 0.79	0.85 / 0.96	0.53 / 0.82
Re=8000 steered st8k / conv rs8kkl	0.97 / 0.84	0.67 / 0.66	0.99 / 1.01	0.56 / 0.67

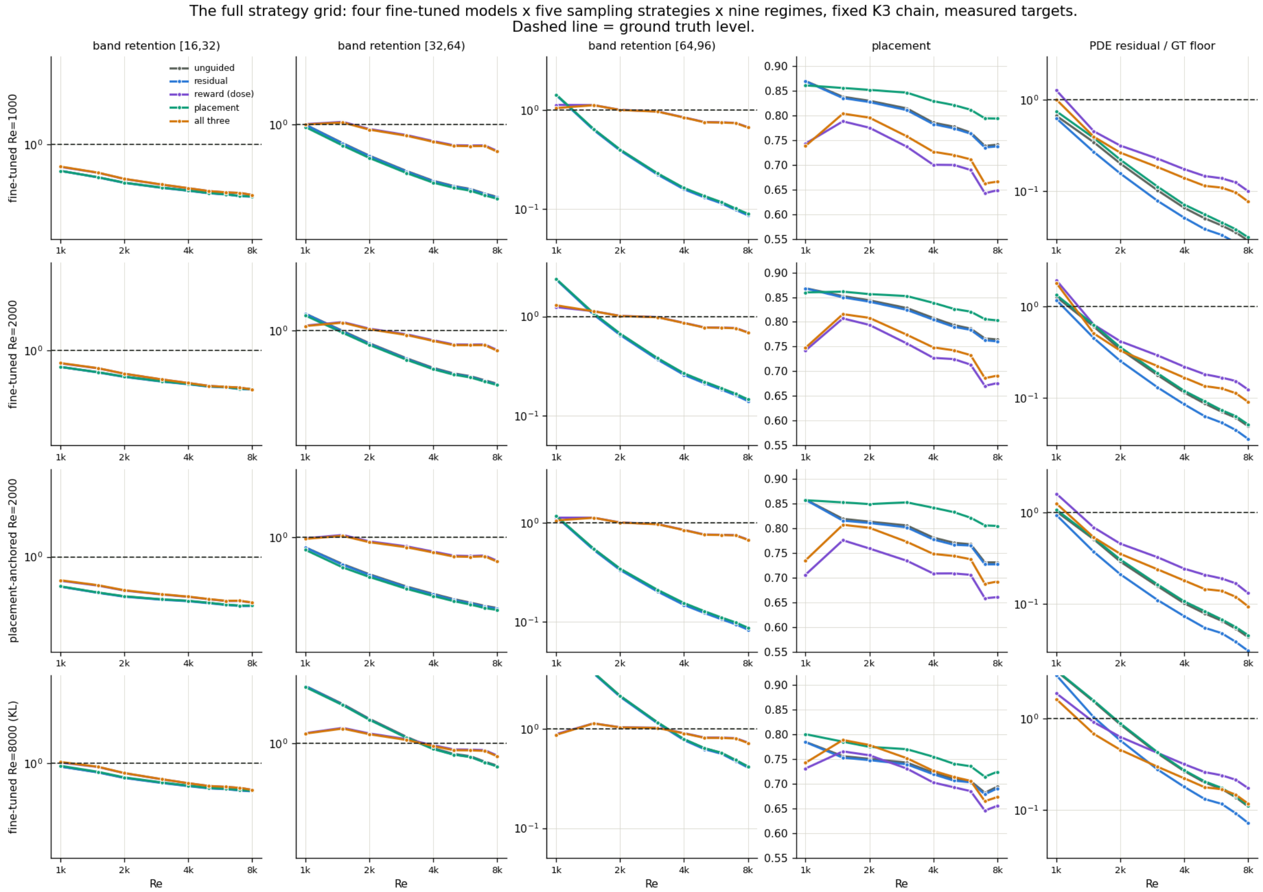
Steered training pays only far from the base regime (+0.12 at Re=8000, large scales pristine) and damages near it: the Re=2000 steered model drains its resolved scales (low-k 0.85 everywhere, the temp-3.5 flaw) and both near models are dial-dependent for no gain. Mechanism: far from base the dial's in-training action is a healthy mean shift; near base it is per-sample variance-squeezing, baked into the weights. Doctrine: bake the dial in for far-regime specialists only. AUTOv2: the fine-band meter fixes the one under-dial (Re=1000: 1.20 to 1.08) with all other choices unchanged.

2b10 · The low-mid term with a calibrated onset (v3)



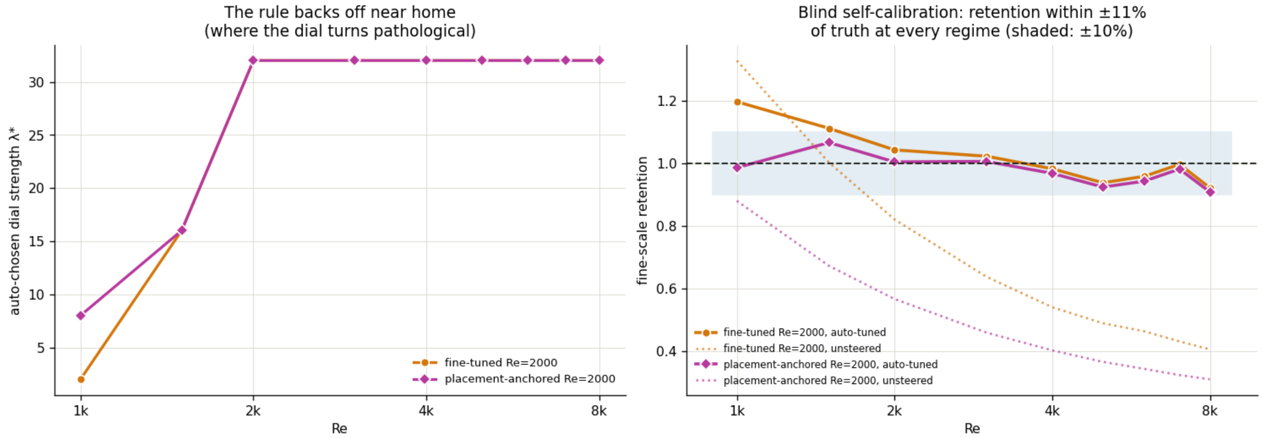
*The roll-off starts at k~6-8 (the observation filter), and [5,16) holds 34% of the energy. A third term d[k_start,16) with the onset detected per model/regime from the unguided spectrum vs the anchor (cold family 6-11, hot family 8-16 - model-dependent, not regime-dependent) lifts [6,16) +0.02-0.04 at zero placement cost, mid/fine bands untouched, and lowers the residual OOD; in-regime it raises it (the in-regime pathology). Production OOD dial: 0.5*d[1,96) + 3*d[k_start,16) + 3*d[16,32) + 3*d[32,96).*

2c · The full strategy grid: 4 models x 5 strategies x 9 regimes



180 cells, fixed K3, measured targets. Deep bands: only the dose gradient holds 0.7–1.0 where everything else collapses to 0.1–0.4; the same dial regulates the hot Re=8000 specialist DOWNWARD at low Re. Placement dial: best placement curve nearly everywhere, free. Honest blind spot: the [16,32) mid-band sags under every strategy (the dose loss under-weights it 6x) — v2 refinement identified.

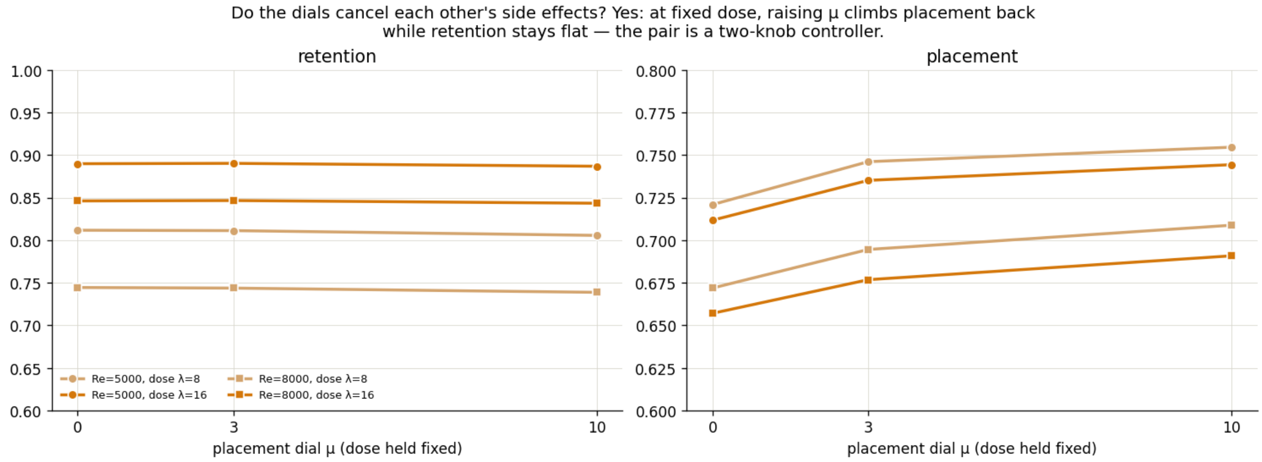
2d · Blind auto-calibration



model	1000	1500	2000	3000	4000	5000	6000	7000	8000
ft-Re2000 λ*	2	16	32	32	32	32	32	32	32
ret @ λ*	1.20	1.11	1.04	1.02	0.98	0.94	0.96	1.00	0.92
place-anch λ*	8	16	32	32	32	32	32	32	32
ret @ λ*	0.99	1.07	1.00	1.01	0.97	0.92	0.94	0.98	0.91

λ chosen from a 32-sample probe against target statistics only; the graded result stays within ±11% of truth at every regime. The rule backs off near home, exactly where the per-sample loss turns variance-squeezing.*

6b · The cancellation accounting (all three dials at once)



	place, unsteered	dose only (λ=16)	+ placement μ=10	refunded	ret cost
Re=5000	0.790	0.712 (–0.078)	0.744 (–0.046)	41%	0.003
Re=8000	0.760	0.657 (–0.103)	0.691 (–0.066)	36%	0.003

Partial, saturating cancellation: ~40% of the placement loss bought back for nothing; the remainder is bounded by the input map's own correlation with truth.

0b · Lineage and the gradient ladder (placed last here; §0b on the web page)

All guidance here is diffusion-posterior-sampling mechanics: subtract a loss gradient inside the reverse step. Shu et al. 2023 (the base model's paper) differentiate the residual at the NOISY iterate (their Linear variant; their main method is classifier-free guidance with the residual as an input feature) — both arms measured inert for dose and placement here, consistent with their own L2-unchanged results. DiffusionPDE (Huang et al. 2024) evaluates losses at the clean estimate and differentiates through the network (strict DPS, ~2–3x cost), guiding toward sparse point observations OF the ground-truth instance — sparse but full-band, an in-distribution spatial gap. Our dials evaluate at the clean estimate and differentiate the LOSS ONLY (no network backward, ~10–15% cost): the observation here is dense but band-limited, so the gap is SPECTRAL, and no observation guidance can fill it — with an out-of-regime prior the missing scales return with the prior's statistics. The three dials factorize the three available information sources: the equation (physics), the instance's coarse observation (placement), and the regime's ensemble statistics (dose) — the third is the novel, sample-free one, and the only one that closes the spectral gap.

Provenance: steering_pilot / steering_measured / steering_full_grid npz stores; trajectory_embed + fields (3 regimes, 5 variants, shared noise); frozen recipe checkpoints; measured_train anchors throughout. Error-bar seed repeats and the steered-training pilot land next and will be folded in.